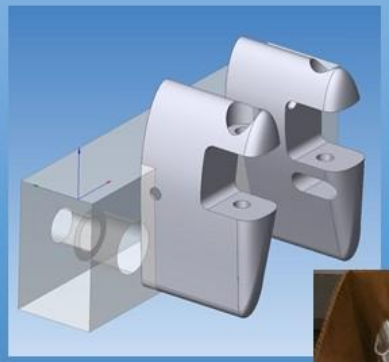
Lessons Learned
and
Murphy's Corollary

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Science and Engineering





Art and Craftsmanship



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Production Test Flying – Normal Procedure





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Test Flown Gliders In The Landing Area



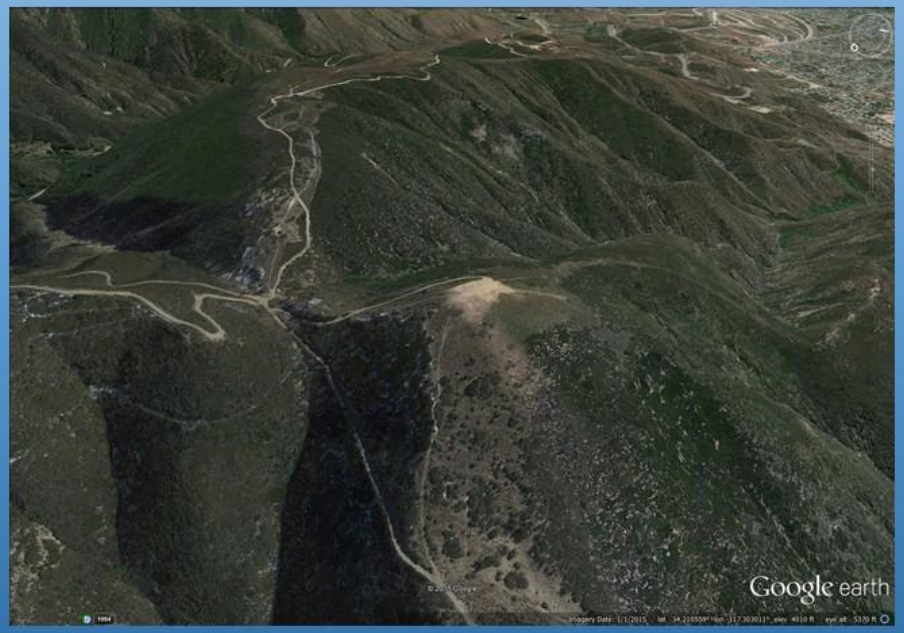
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Production Test Flying – Top Landing

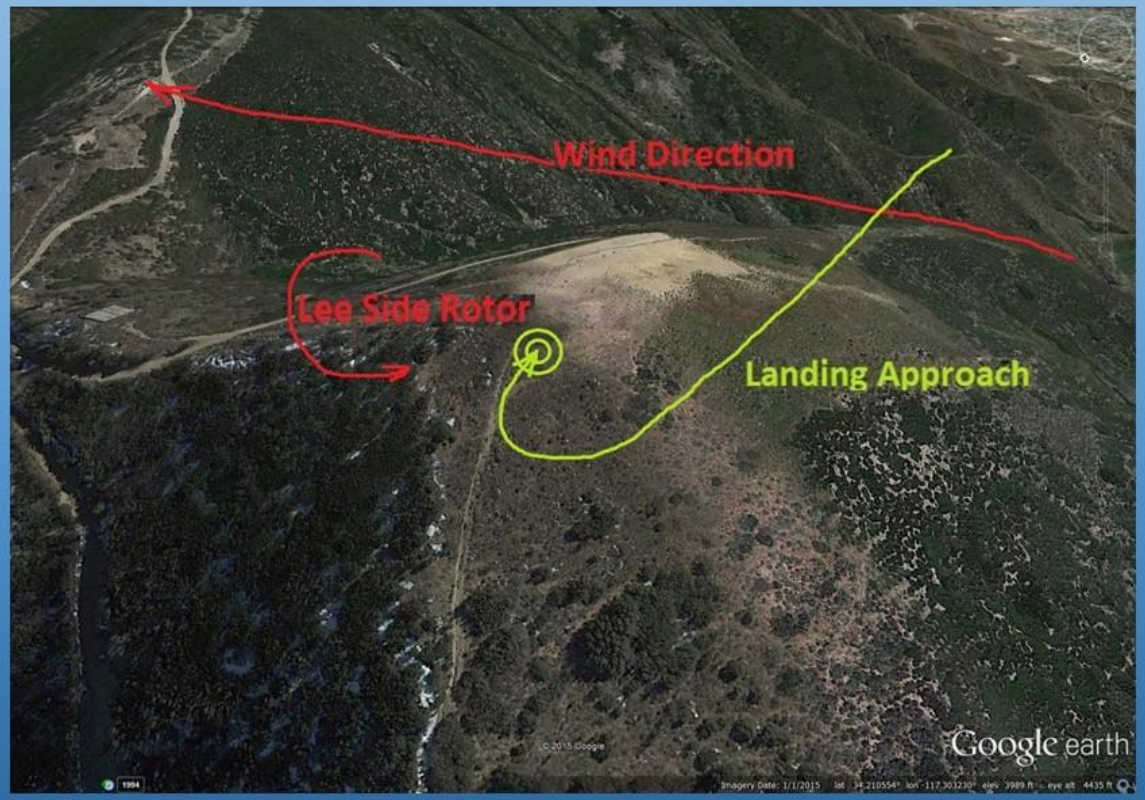


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Marshall Peak, San Bernardino



Marshall Peak Top Landing Approach



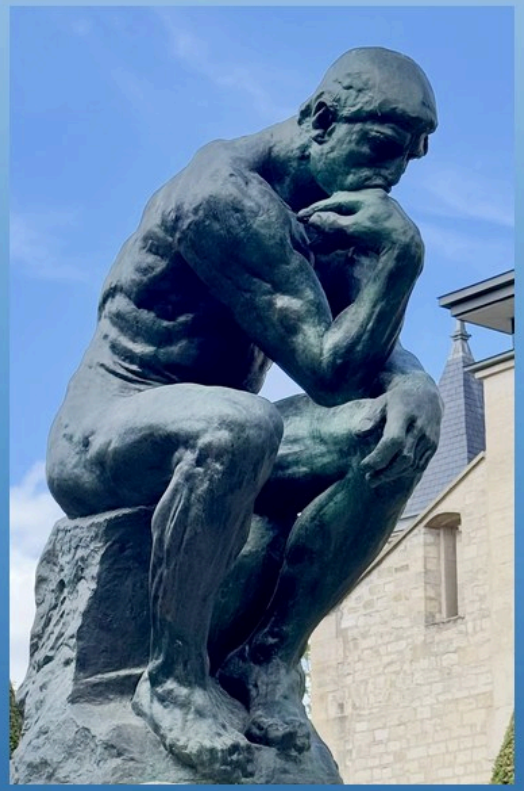












Quality of decisions determines safety

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The go decision requires a very high degree of confidence in a positive outcome.

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Feedback Loop



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Feedback Loop With Lowered Evaluative Criteria



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Murphy's Law:

Anything that can go wrong, will go wrong.

Murphy's Corollary

Anything that did not go wrong could not have gone wrong.

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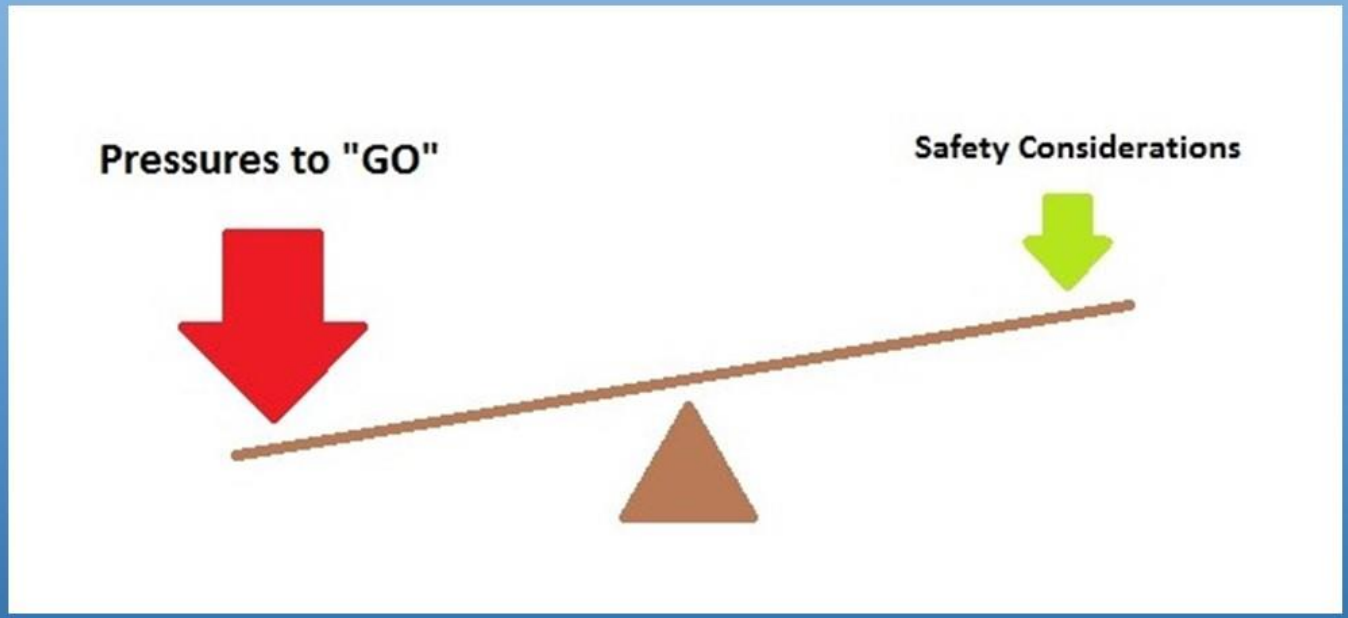
$$.99^{18} = .83$$

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$$1 - .83 = .17 = 1/6$$

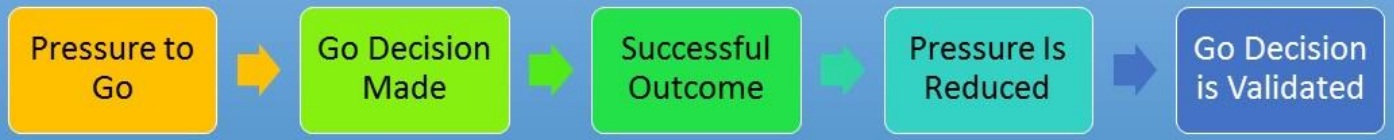
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Pressure To Make The Go Decision



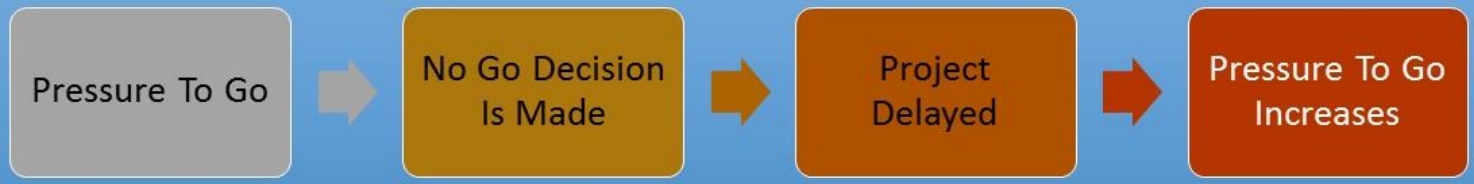
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Go Decision Reduces Pressure



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No Go Decision Increases Pressure



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1. Apollo 1



2. Space Shuttle Challenger



3. Space Shuttle Columbia



1. Apollo 1 - 1967

2. Space Shuttle Challenger - 1986

3. Space Shuttle Columbia - 2003

The Challenge

- Deceptive Mathematics Of Probability
- Self-Deceptive Psychology Of Evaluating Our Own Decisions
- Inevitable Pressure To Make The "Go" Decision

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Continual Awareness

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Expand The Scope
Of Lessons Learned

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Questions?